SANJIVANI UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

At: kopargaon, Dist:Ahmednagar, Maharashtra, India – 423601

DEPARTMENT (Artificial Intelligence and Machine Learning)

COURSE: DevOps and Cloud (IBM)

(24UAMAE201)

Academic Year 2026-2027

NAME	**Vaishnavi Jadhav**
PRN NO	**2125UMLF1100**
SEM/YEAR	**2nd Year/3rd Sem**
BATCH	**SY(A)**

SUBMITTED TO

Mr. Karan Gaikwad Sir
(Professor)

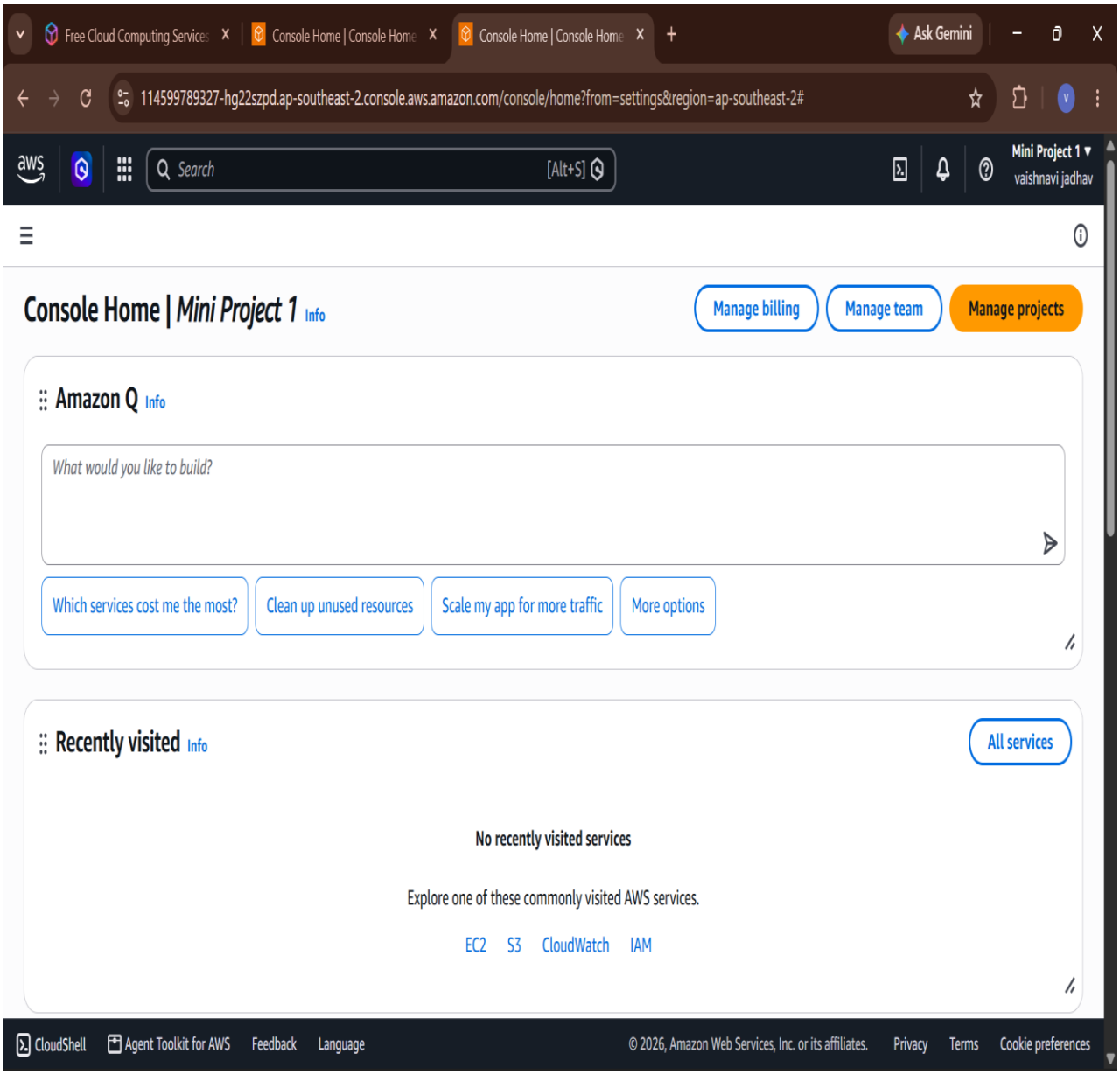
SUBMITTED BY

Vaishnavi Jadhav
(B.Tech 23nd year)

PRN: 2125UMLF1100

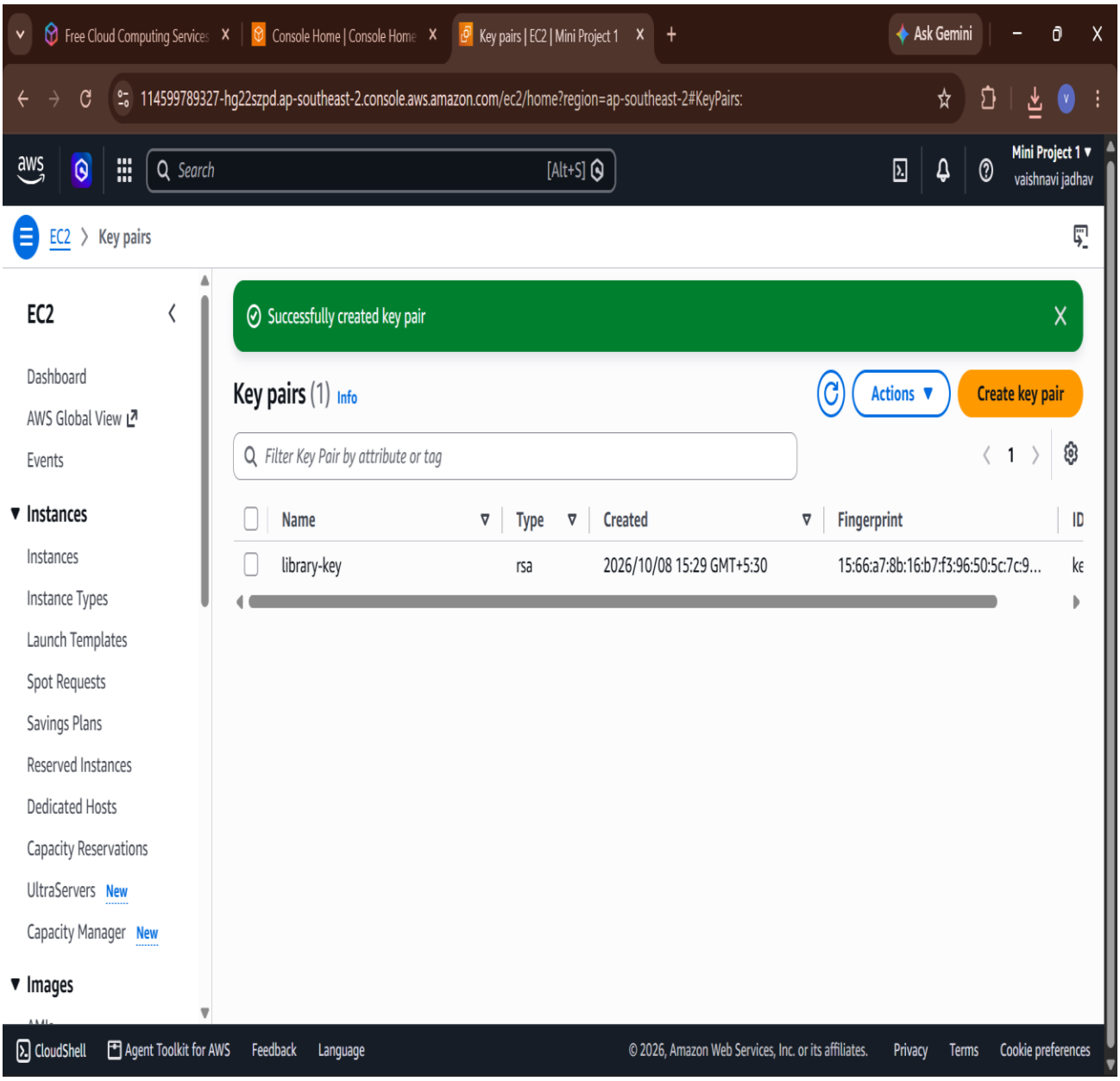
Step 1: Log in and select the region

- The AWS Management Console after login, with the region set to **Asia Pacific (Sydney)**. All resources for the **Library Management Platform** are created in this region.



Step 2: Create a key pair

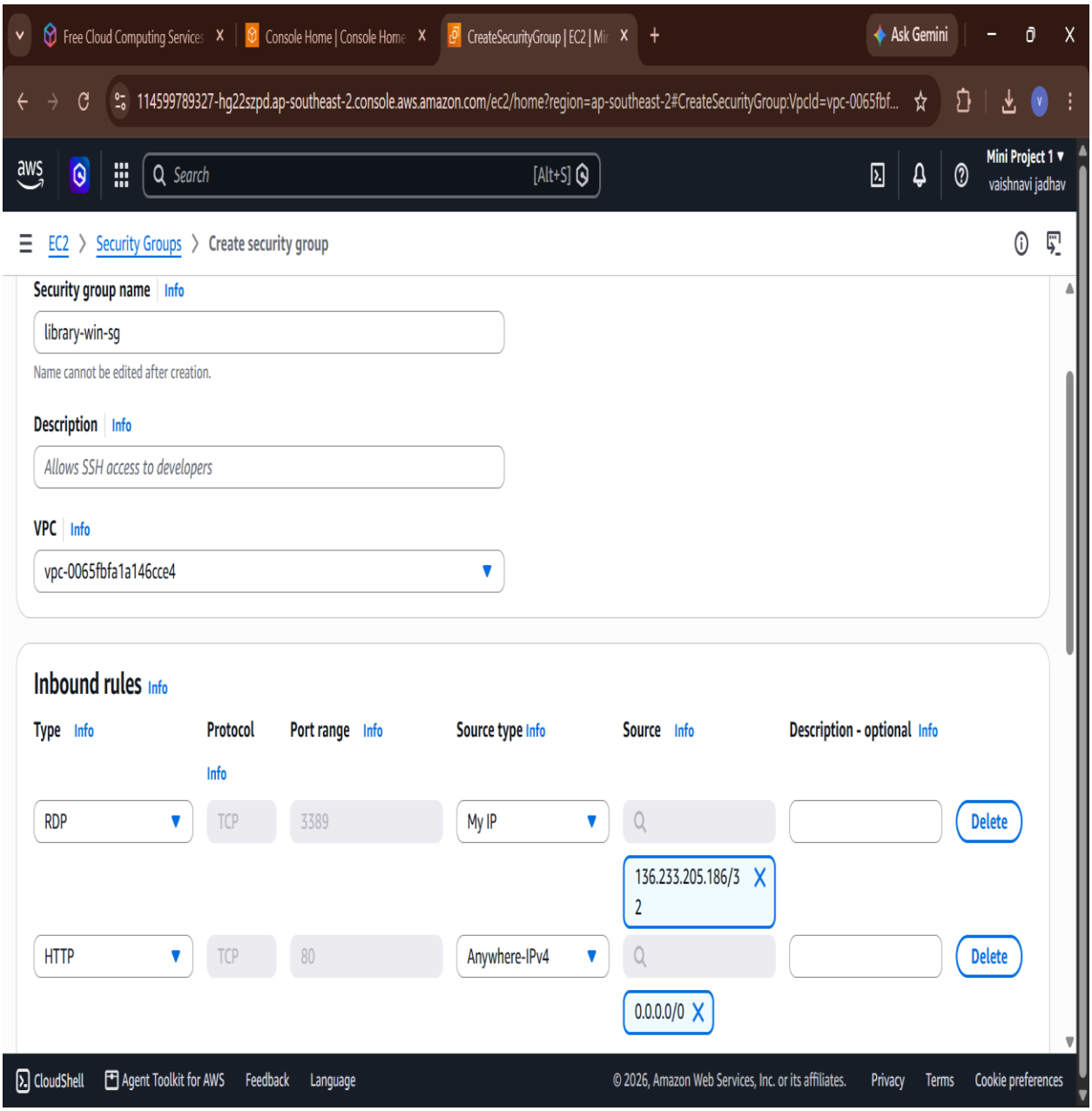
- A key pair named library-key was created. For Windows servers, the private key is used to decrypt the Administrator password needed for Remote Desktop login.



Step 3: Create the security group

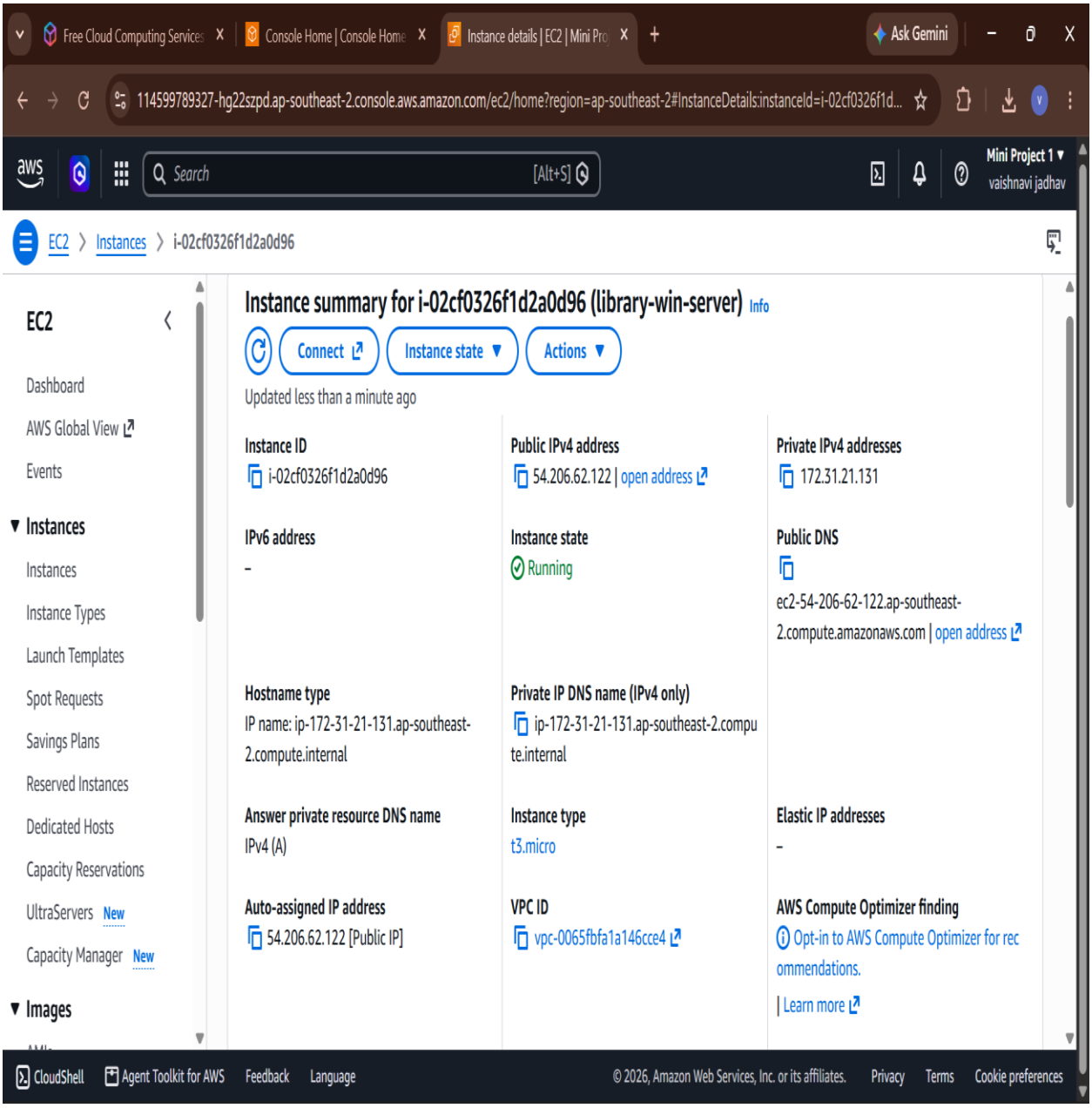
- The inbound rules tab of `library-win-sg:`

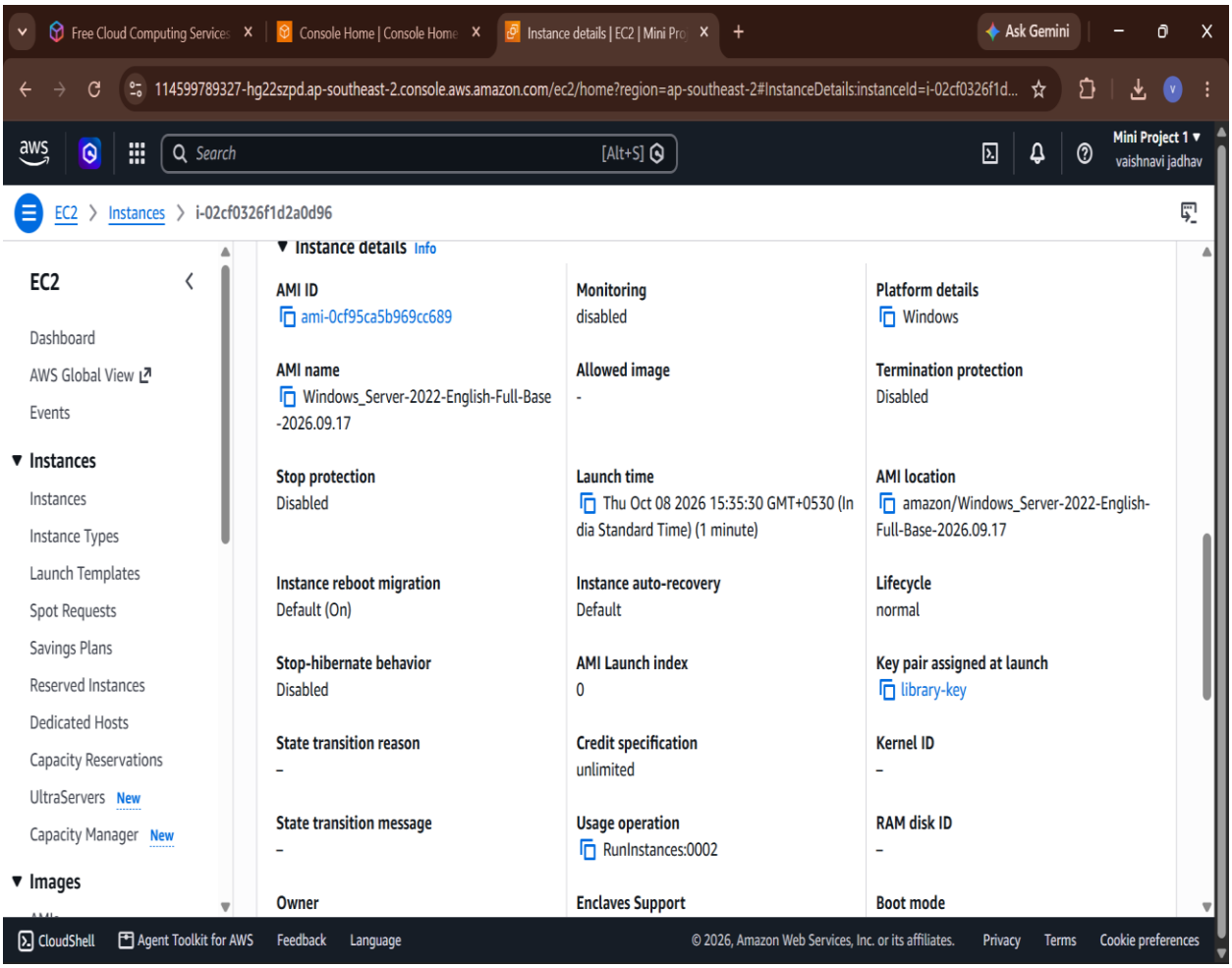
Security group works as a virtual firewall. It allows RDP (3389) only from my IP for secure administration and HTTP (80) from anywhere so users can access the library website.



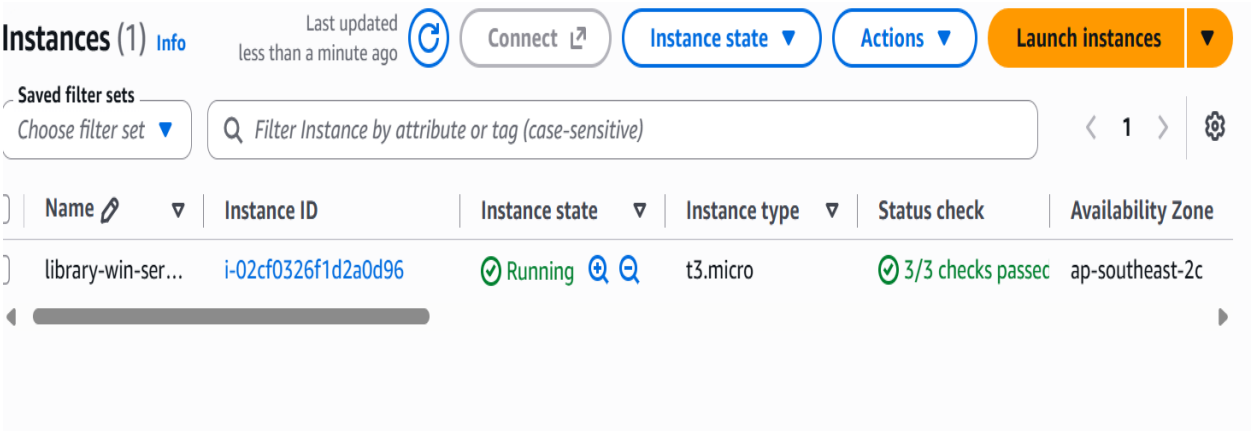
Step 4: Launch the Windows EC2 instance

- A Windows Server 2022 EC2 instance named library-win-server is configured with the key pair and security group created earlier. It hosts the application tier of the platform.



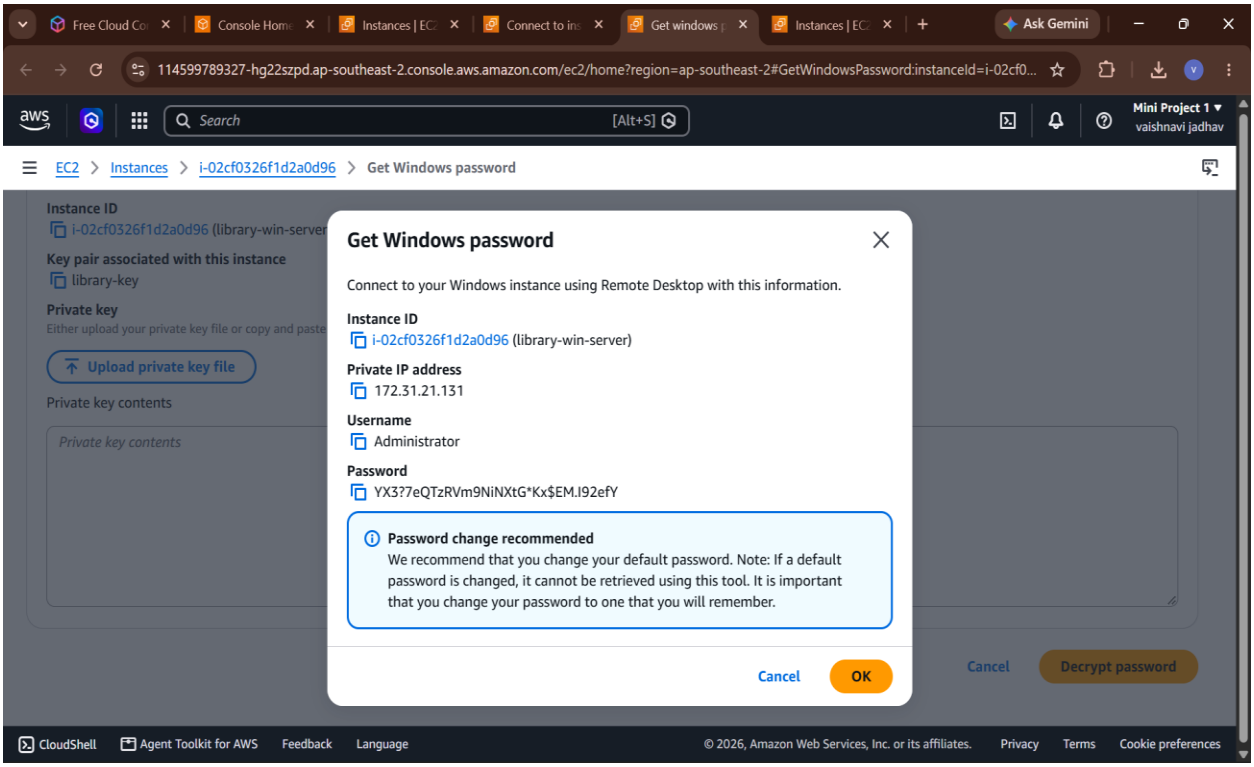


- The Windows instance is running and all status checks have passed. Its public IP address is used for both Remote Desktop and the website.

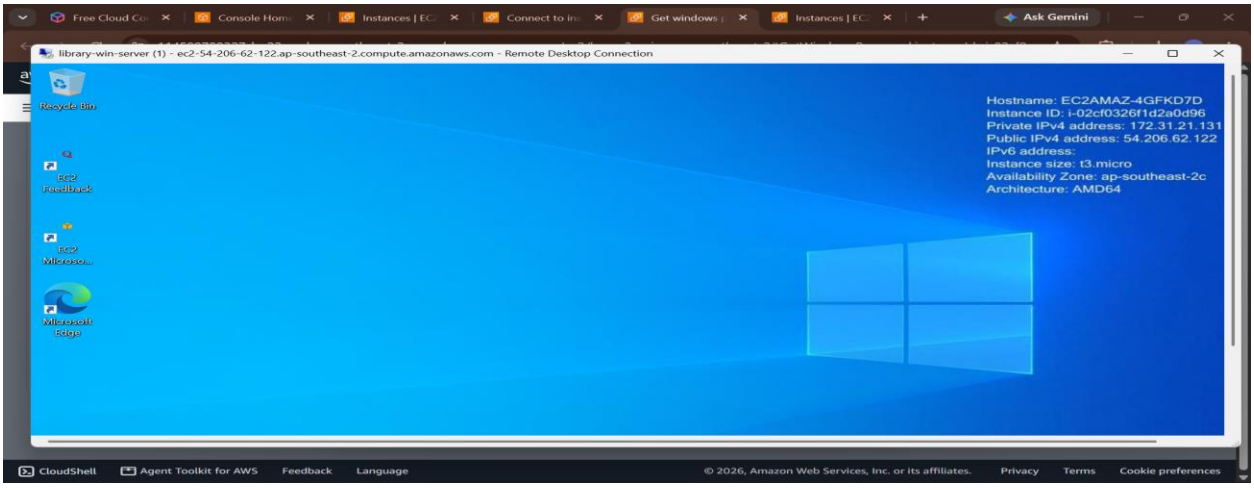


Step 5: Get the password and connect with Remote Desktop

- The Administrator password was retrieved by decrypting it with the private key. This ensures only the key holder can access the server."

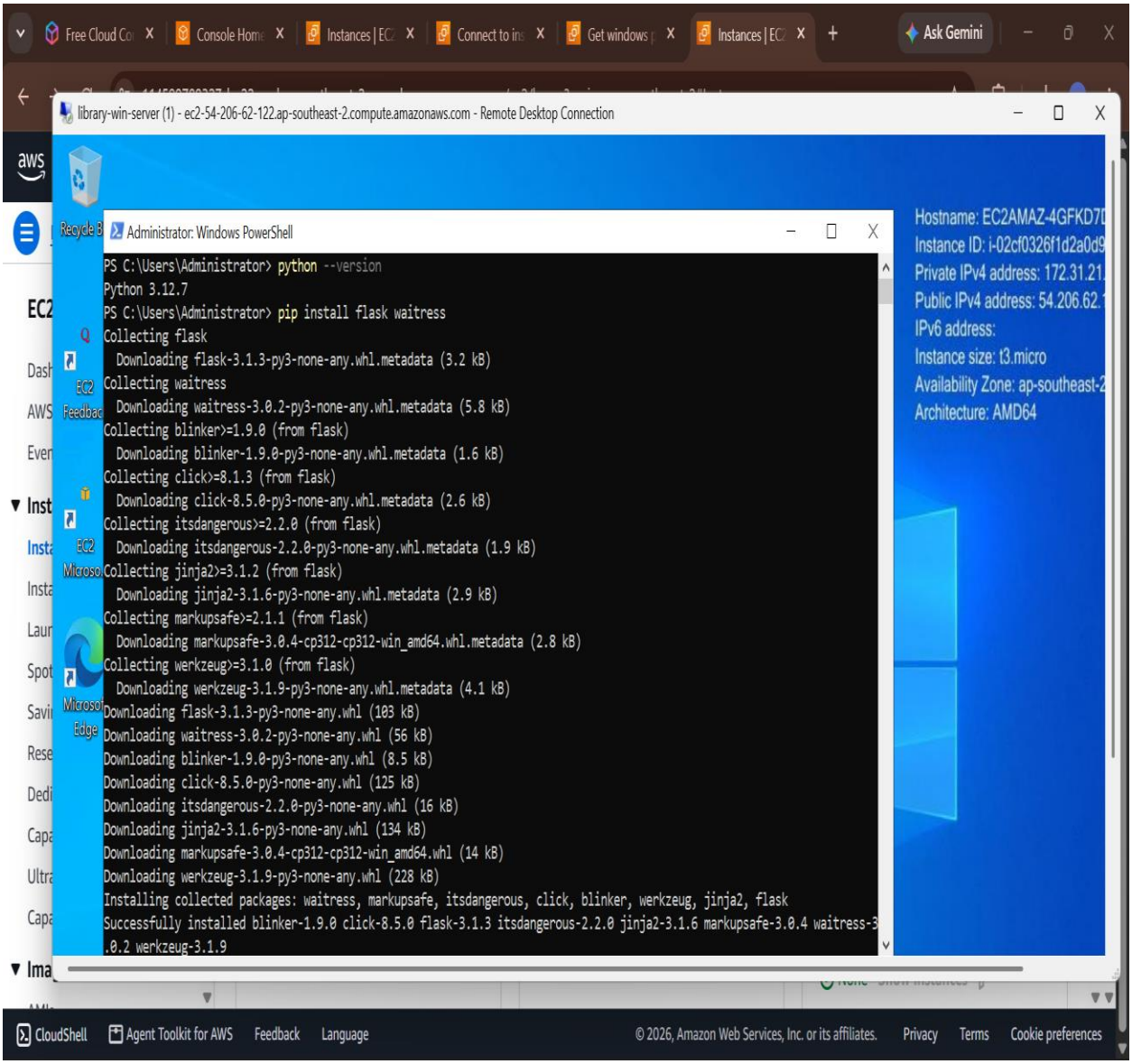


- Successfully connected to the AWS Windows Server using Remote Desktop. All further setup is done on this server.



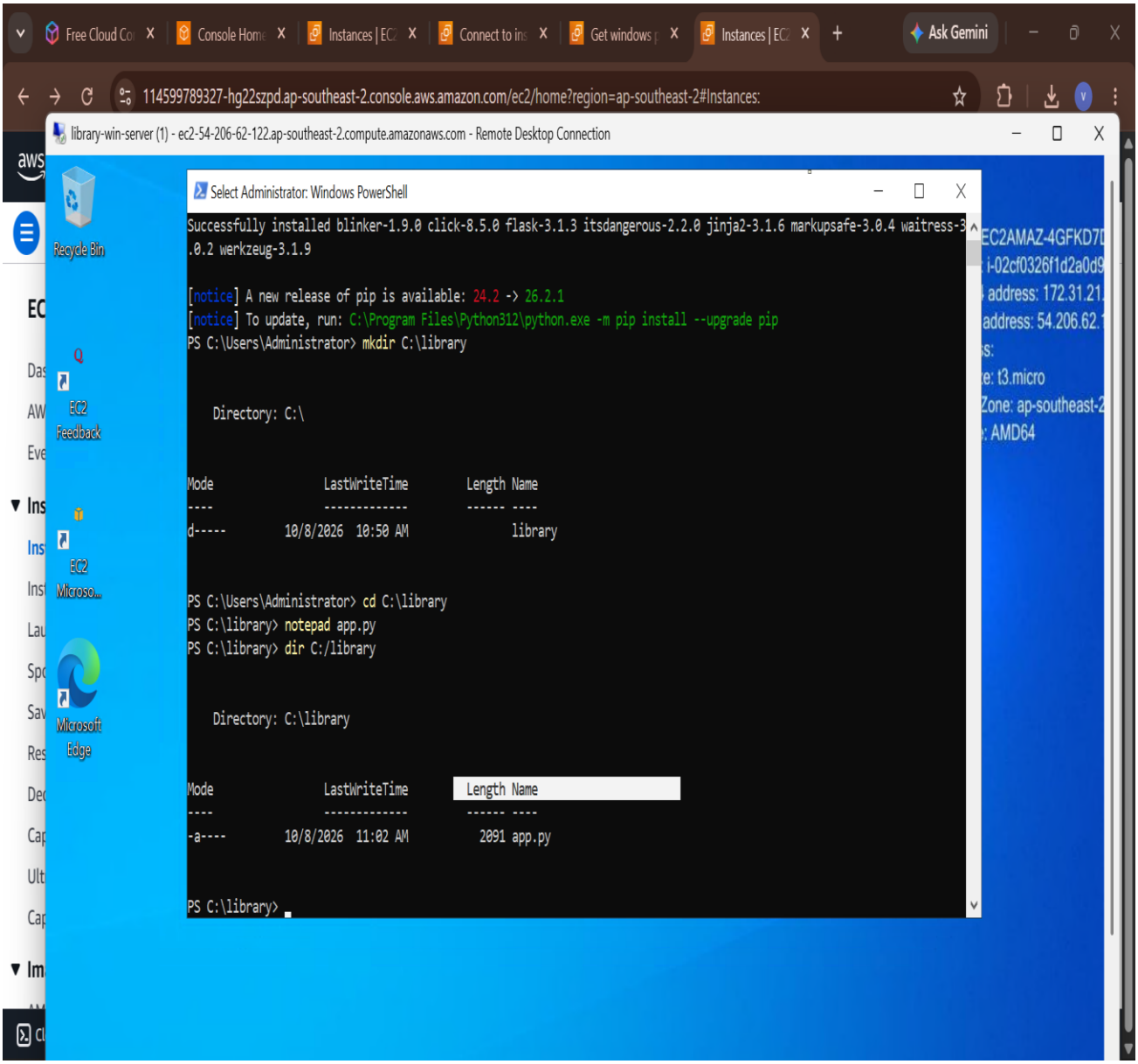
Step 6: Install Python on the server

- Python and the Flask web framework were installed on the server. Waitress is the production web server used to run the application on Windows.



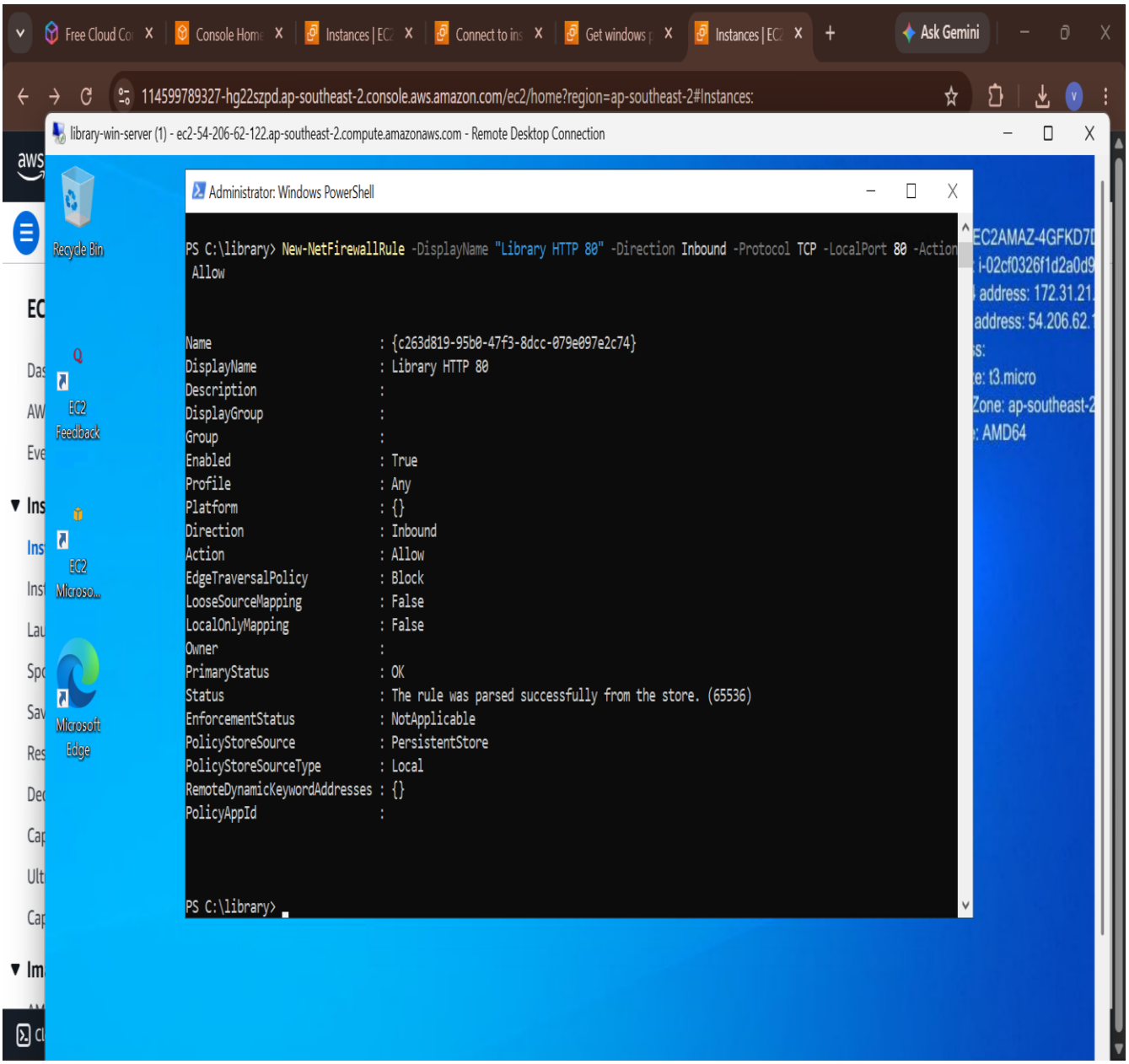
Step 7: Create the application

- The application source code was created in C:\library. It implements book search, issue with a 14-day due date, and return with automatic fine calculation at Rs 5 per overdue day, as defined in the MVP scope.



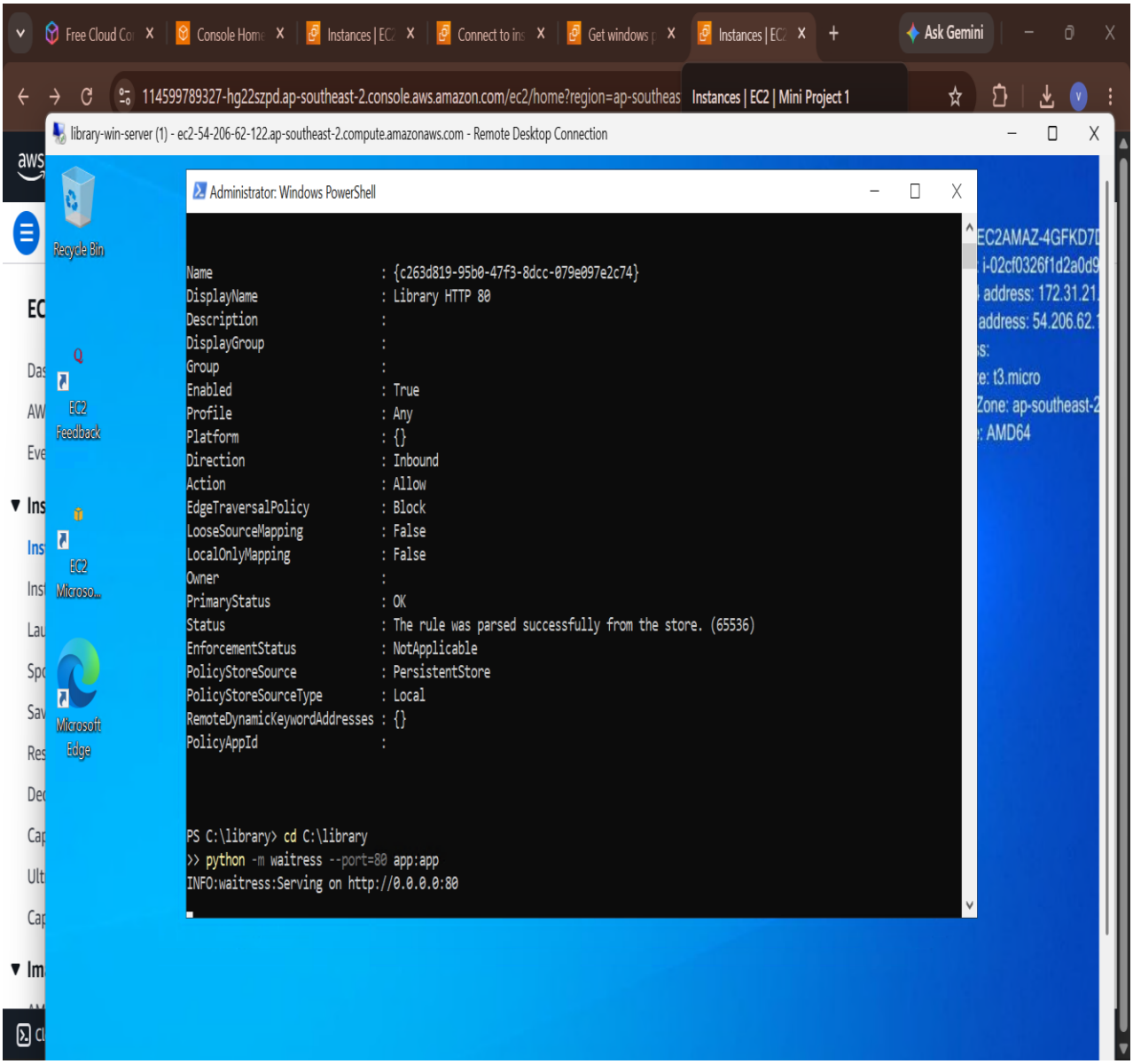
Step 8: Open port 80 in Windows Firewall

- An inbound rule was added to Windows Firewall to allow HTTP traffic on port 80. Both the AWS security group and the OS firewall must allow the port for the site to be reachable.



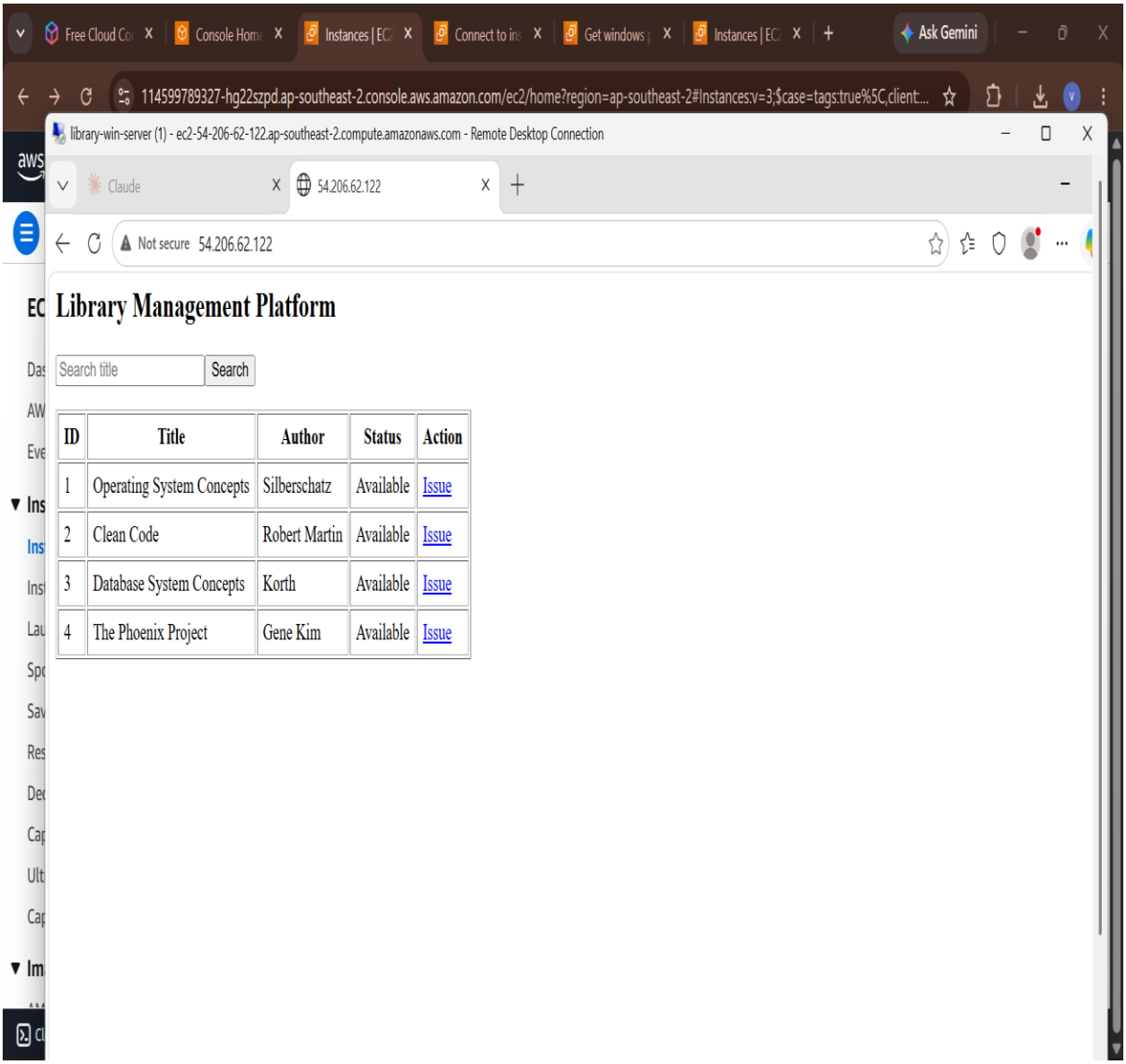
Step 9: Run the application

- The Library Management Platform is running on the Windows server through the Waitress web server, listening on port 80.

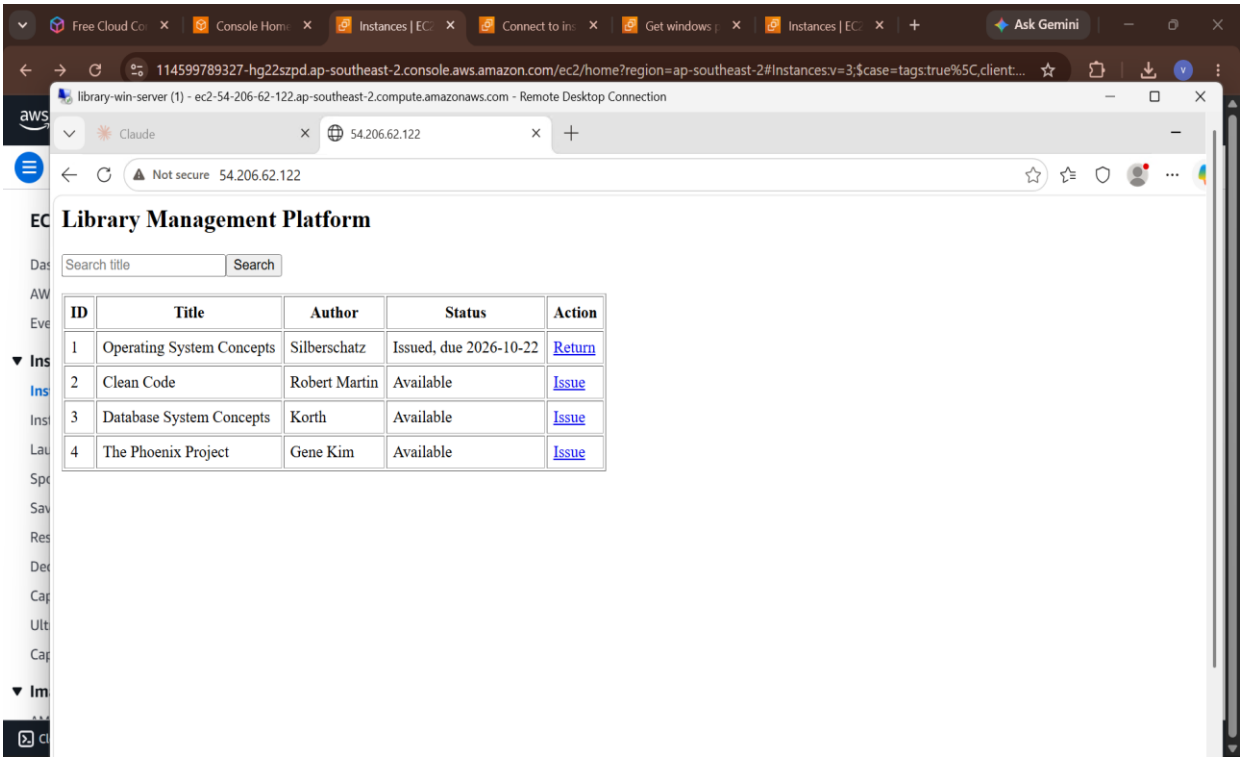


Step 10: Open the website

- The Library Management Platform running live on AWS Windows Server, showing the catalog with availability status.



- Issuing a book updates its status and sets a due date 14 days ahead, demonstrating the circulation engine requirement (FR-07).



- Returning a book makes it available again and calculates the fine automatically, demonstrating the fine management requirement.

